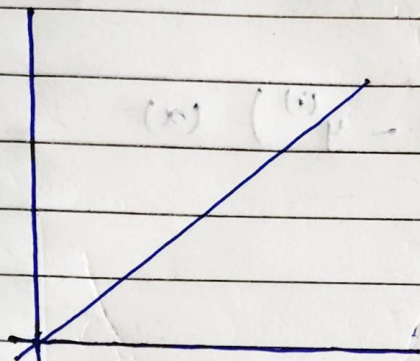


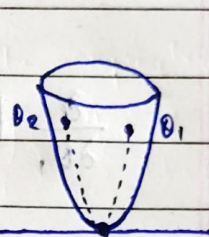
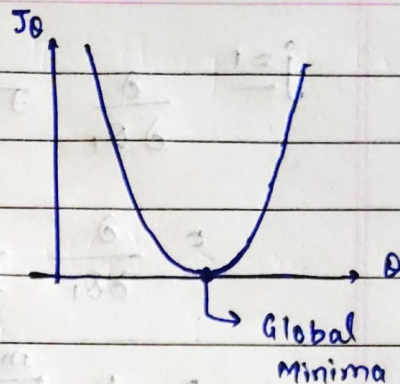
Final conclusion:

GRADIENT DESCENT



$$\theta_0 = 0$$

$\theta_1 = \text{changeable}$



Convergence Algorithm.

repeat until convergence

{

$$\theta_j = \theta_j - \alpha \left[\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) \right] \quad j=0 \text{ and } 1$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_j} \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

if

$$j=0 \Rightarrow \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$h_{\theta}(x) = \theta_0 + \theta_1 x \rightarrow 0$$

$$= \frac{\partial}{\partial \theta_0} \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 \right]$$

$$= \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \times 1$$

Note: Differentiate with θ_0 so θ_1 is constant

$$\Rightarrow \theta_1 x = 0 \quad \text{and} \quad \theta_0 = 1$$

$$j=1 \quad \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$= \frac{\partial}{\partial \theta_1} \frac{1}{2m} \left[\sum_{i=1}^m (1\theta_0 + \theta_1 x) - y^{(i)} \right]^2$$

$$= \frac{1}{m} \sum_{i=1}^m (1\theta_0 + \theta_1 x) - y^{(i)} \quad (x)$$

$$\cdot \frac{\partial}{\partial \theta_1} [1\theta_0 + \theta_1 x] \Rightarrow x$$

Repeat until convergence.

$$\left\{ \begin{aligned} \theta_0 &= \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_0(x)^{(i)} - y^{(i)}) \end{aligned} \right\}$$

$$\theta_1 = \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_0(x)^{(i)} - y^{(i)}) x^{(i)}$$

}